

and perhaps a new round of incentives [is] needed to encourage domestic product design companies to start building local cellphones, laptops, desktops, servers, routers and other products for automotive, communications and industrial segments to substitute imports.

We need to encourage local businesses to build domestic companies like Xiaomi, Oppo, Nokia, Huawei, Lenovo, and Dell. These OEMs should be supported by a supply of locally designed chips. Thus, we also need to create fabless design companies in India like MediaTek, Broadcom, Qualcomm and Marvell. And to make it even more challenging, these two sets of activities—designing and building devices, and designing local chips that go into these devices—need to start now, hand in hand, so that we have ample availability of locally designed chips, available in a timely manner to fill our fabs as they get ready for production.

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Q: What are the things that India should be mindful of to not repeat the mistakes made in the past?

A: Frankly, I am not aware of any mistakes that were made by India in the recent past relating to manufacturing chips in India. To the best of my knowledge, never before has the Government of India ever developed such a scheme for encouraging the setting up of a semiconductor fab in India. However, I will be remiss if I failed to share a little-known secret that was shared with me by the late Gurpreet Singh, the founder of Continental Device India Pvt. Ltd (CDIL), a pioneer [in the field of] manufacturing silicon chips and devices [who had been making semiconductors] in Delhi and Faridabad since 1964. CDIL is my first love; this is where I began my career and was the place where I fell in love with semiconductor technology. Not long ago, Gurpreet Singh shared with me that Robert Noyce, the co-founder of Intel, had visited India in 1969 to explore the opportunity of setting up a semiconductor manufacturing facility here. Unfortunately, the government

told him that he could only set up a fab that would not exceed hundreds of thousands of chips, [which is] a complete non-starter. I have often wondered how the tides with Taiwan and China may have turned had we allowed Intel to set up a fab in India back then!

However, there are many things India should keep in mind as it embarks on this vital journey. The semiconductor process tech is not static; the nodes are moving targets. Hence a large amount of capital investment is needed every time the process needs to be moved to the next generation. Private businesses, especially in India, shy away from heavy capital-intensive projects in favour of quick return-on-investment businesses, which are aplenty in the field of software, especially software services. Therefore, the Indian government will have to play a large role in continuing to support the investments for the proliferation of the semiconductor ecosystem and related industries.

Also, if you analyse the \$280 billion the US has recently approved under their CHIPS Act, the bulk of it has been allocated for R&D. This is where the new processes, materials and technologies get innovated for continuing the rapid progress we have made in technology over the past 50 years. ISM has been exploring opportunities to set up an R&D centre to develop intellectual property and catch up with the rest of the world on new chips and packaging technologies.

Last but not least, there is a paucity of semiconductor know-how in India. Even though we have tens of thousands of design engineers currently working with the who's who of the world's top multinational semiconductor companies in India, there are hardly any semiconductor engineers trained in the knowledge of device physics and process technology that is essential for fabricating and manufacturing chips designed by the design engineers. Once again, ISM has already put the spotlight on this issue, and many parallel efforts are in place to start addressing this.

The semiconductor industry started 50 years ago and has been rapidly evolving since then. We are entering it many decades late, so the job of creating an entire new universe of semiconductors is daunting, to say the least. We have no choice but to undertake this enormous challenge for the critical importance it holds for India's future economic prosperity and defence.

This complex and multifaceted task of creating an entire semiconductor ecosystem will require a very long time to get established. To achieve success, we will need to have patience, remain steadfast and consistent, and pursue it relentlessly despite the many ups and downs that will be encountered along the way. **BT**

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